

Submission Worksheet

Submission Data

Course: IT114-003-F2025

Assignment: IT114 Java Problems

Student: Alanor S. (ass89)

Status: Submitted | **Worksheet Progress:** 100+%

Potential Grade: 11.00/10.00 (110.00%)

Received Grade: 0.00/10.00 (0.00%)

Started: 9/25/2025 2:25:44 PM

Updated: 10/1/2025 10:42:30 PM

Grading Link: <https://learn.ethereallab.app/assignment/v3/IT114-003-F2025/it114-java-problems/grading/ass89>

View Link: <https://learn.ethereallab.app/assignment/v3/IT114-003-F2025/it114-java-problems/view/ass89>

Instructions

- Overview Link: <https://youtu.be/Mrahk6SEYao>
- 1. Ensure you read all instructions and objectives before starting.
- 2. Create a new branch from `main` called `M2-Homework`
 1. `git checkout main` (ensure proper starting branch)
 2. `git pull origin main` (ensure history is up to date)
 3. `git checkout -b M2-Homework` (create and switch to branch)
- 3. Copy the template code from here: [GitHub Repository - M2 Homework](#)
 - It includes Problems 1-4 and a `BaseClass`. Put all into an `M2` folder or similar (adjust package reference at the top if you chose a different folder name).
 - Immediately record to history
 - `git add .`
 - `git commit -m "adding M2 HW baseline files"`
 - `git push origin M2-Homework`
 - Create a Pull Request from `M2-Homework` to `main` and keep it open
- 4. Fill out the below worksheet
 - Each Problem requires the following as you work
 - Ensure there's a comment with your UCID, date, and brief summary of how the problem was solved
 - Initial outline/plan of how you'll solve it via comments (add/commit after this stage)
 - Code solution (add/commit periodically as needed)
- 5. Once finished, click "Submit and Export"
- 6. Locally add the generated PDF to a folder of your choosing inside your repository folder and move it to Github
 1. `git add .`
 2. `git commit -m "adding PDF"`
 3. `git push origin M2-Homework`
 4. On Github merge the pull request from `M2-Homework` to `main`
- 7. Upload the same PDF to Canvas
- 8. Sync Local
 1. `git checkout main`

Section #1: (2 pts.) Problem 1 - Odds

Progress: 100%

≡ Task #1 (2 pts.) - Edit the `printOdds` method to output odd values of the array

Progress: 100%

Details:

- Only make edits where noted via provided comments
- Challenge: Print odd values only in a single line separated by commas
- Step 1: sketch out plan using comments (include ucid and date)
- Step 2: Add/commit your outline of comments (required for full credit)
- Step 3: Add code to solve the problem (add/commit as needed)

Part 1:

Progress: 100%

Details:

Two screenshots are expected

1. Snippet of relevant code showing solution (with ucid/date comment)
2. Full output of executing the program

```

1  BaseClass.java  Problem1.java  Problem2.java  Problem3.java  Problem4.java
2  > Problem1.java  > Problem1: 50 printOdds(int[] arr)
3  package edu.cmu.cs;
4
5  public class Problem1 extends BaseClass {
6
7      private static int[] array1 = {1,2,3,4,5,6,7,8,9,10};
8      private static int[] array2 = {0,1,2,3,4,5,6,7,8,9,10};
9      private static int[] array3 = {0,1,2,3,4,5,6,7,8,9,10};
10     private static int[] array4 = {0,1,2,3,4,5,6,7,8,9,10};
11
12     // Challenge: Print odd values only in a single line separated by commas
13     // Step 1: sketch out plan using comments (include ucid and date)
14     // Step 2: Add/commit your outline of comments (required for full credit)
15     // Step 3: Add code to solve the problem (add/commit as needed)
16     // UCID: [your ucid] Date: [your date]
17     // NAME: [your name]
18
19     public void printOdds(int[] arr, int arrLength) {
20         // Challenge: Print odd values only in a single line separated by commas
21         // Step 1: sketch out plan using comments (include ucid and date)
22         // Step 2: Add/commit your outline of comments (required for full credit)
23         // Step 3: Add code to solve the problem (add/commit as needed)
24         // UCID: [your ucid] Date: [your date]
25         // NAME: [your name]
26
27         // Print odd values only in a single line separated by commas
28         for (int i = 0; i < arrLength; i++) {
29             if (arr[i] % 2 == 1) {
30                 System.out.print(arr[i] + ", ");
31             }
32         }
33         System.out.println();
34     }
35 }

```

Part 1 of the code

```

1  BaseClass.java  Problem1.java  Problem2.java  Problem3.java  Problem4.java
2  > Problem1.java  > Problem1: 50 printOdds(int[] arr)
3  package edu.cmu.cs;
4
5  public class Problem1 extends BaseClass {
6
7      private static int[] array1 = {1,2,3,4,5,6,7,8,9,10};
8      private static int[] array2 = {0,1,2,3,4,5,6,7,8,9,10};
9      private static int[] array3 = {0,1,2,3,4,5,6,7,8,9,10};
10     private static int[] array4 = {0,1,2,3,4,5,6,7,8,9,10};
11
12     // Challenge: Print odd values only in a single line separated by commas
13     // Step 1: sketch out plan using comments (include ucid and date)
14     // Step 2: Add/commit your outline of comments (required for full credit)
15     // Step 3: Add code to solve the problem (add/commit as needed)
16     // UCID: [your ucid] Date: [your date]
17     // NAME: [your name]
18
19     public void printOdds(int[] arr, int arrLength) {
20         // Challenge: Print odd values only in a single line separated by commas
21         // Step 1: sketch out plan using comments (include ucid and date)
22         // Step 2: Add/commit your outline of comments (required for full credit)
23         // Step 3: Add code to solve the problem (add/commit as needed)
24         // UCID: [your ucid] Date: [your date]
25         // NAME: [your name]
26
27         // Print odd values only in a single line separated by commas
28         for (int i = 0; i < arrLength; i++) {
29             if (arr[i] % 2 == 1) {
30                 System.out.print(arr[i] + ", ");
31             }
32         }
33         System.out.println();
34     }
35 }
36
37 // Main method
38 public static void main(String[] args) {
39     Problem1 p1 = new Problem1();
40     p1.printOdds(array1, array1.length);
41     p1.printOdds(array2, array2.length);
42 }

```

Part 2 of the code

```
Dear@DESKTOP-MPGCNEP MINGW64 ~/Desktop/NJIT/IT114/asa89-IT114-003 (M2-Homework)
$ java M2.Problem1
Running Problem 1 for [asa89] [2025-09-28T17:59:32.274268300]
Objective: Print out only odd values in a single line separate by commas
Problem 1: Original Array: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
Output Array: 1,3,5,7,9

Problem 2: Original Array: [9, 8, 7, 6, 5, 4, 3, 2, 1, 0]
Output Array: 9,7,5,3,1

Problem 3: Original Array: [8, 8, 1, 1, 2, 2, 3, 3, 4, 4, 5, 5, 6, 6, 7, 7, 8, 8, 9, 9]
Output Array: 1,1,3,3,5,5,7,7,9,9

Problem 4: Original Array: [9, 9, 8, 8, 7, 7, 6, 6, 5, 5, 4, 4, 3, 3, 2, 2, 1, 1, 0, 0]
Output Array: 9,9,7,7,5,5,3,3,1,1

Completed Problem 1 for [asa89] [2025-09-28T17:59:32.314550300]
Dear@DESKTOP-MPGCNEP MINGW64 ~/Desktop/NJIT/IT114/asa89-IT114-003 (M2-Homework)
$ []
```

Output of the execution of the code

 Saved: 10/1/2025 1:02:11 PM

Part 2:

Progress: 100%

Details:

Direct link to the file in the homework related branch from Github (should end in `.java`)

URL #1

<https://github.com/AICSS442/asa89-IT114-003-M2-Homework/M2/Problem1.java>



URL

<https://github.com/AICSS442/asa>

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Part 3:

Progress: 100%

Details:

Briefly explain `how` the code solves the challenge (note: this isn't the same as `what` the code does)

Your Response:

I solved the challenge by filtering the original array to keep only the odd numbers and then printing them in order, separated by commas. I first collected the odd numbers into a separate list, which made it easy for me to format the output correctly with commas between the values. So essentially, I focused on isolating the numbers I needed and controlling the output format to solve it.

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Section #2: (2 pts.) Problem 2 - Sum

Progress: 100%

Task #1 (2 pts.) - Edit the `sumValues` method to sum the

Progress: 100%

- Only make edits where noted via provided comments
- Challenge 1: Sum all the values of the passed in array and assign to `total`
- Challenge 2: Have the sum be represented as a number with exactly 2 decimal places
- Example: 0.1 would be shown as 0.10, 1 would be shown as 1.00, etc
- Step 1: sketch out plan using comments (include uid and date)
- Step 2: Add/commit your outline of comments (required for full credit)
- Step 3: Add code to solve the problem (add/commit as needed)

Progress: 100%

1. Snippet of relevant code showing solution (with uid/date comment)
2. Full output of executing the program

Part 1 codePart 2 code

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int n = scanner.nextInt();
        int[] arr = new int[n];
        for (int i = 0; i < n; i++) {
            arr[i] = scanner.nextInt();
        }
        int total = 0;
        for (int i = 0; i < n; i++) {
            total += arr[i];
        }
        System.out.println(total);
    }
}
```

Output of the execution of the code

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Part 2:

Progress: 100%

Details:

Direct link to the file in the homework related branch from Github (should end in `.java`)

URL #1

<https://github.com/AlCSS442/asa89-IT114-1003M2-Homework/M2/Problem2.java>



URL

<https://github.com/AlCSS442/asa>

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Part 3:

Progress: 100%

Details:

Briefly explain `how` the code solves the challenges (note: this isn't the same as `what` the code does)

Your Response:

I solved the first challenge by iterating through each array and summing all the values into a total. For the second challenge, I made sure the total was displayed with exactly two decimal places by rounding and formatting it as a string. This way, even numbers that normally have more or fewer decimals, like 0.1 or 1, can appear as 0.10 or 1.00.

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Section #3: (2 pts.) Problem 3 - Conversion

Progress: 100%

Task #1 (2 pts.) - Edit the `bePositive` method to make each value positive, convert it back to the original data type, and set it to the proper slot in the `output` array

Progress: 100%

Part 3 code

```
import java.util.*;
import java.math.*;

public class Problem3 {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int n = scanner.nextInt();
        int m = scanner.nextInt();
        int k = scanner.nextInt();
        int[] arr1 = new int[n];
        int[] arr2 = new int[m];
        for (int i = 0; i < n; i++) {
            arr1[i] = scanner.nextInt();
        }
        for (int i = 0; i < m; i++) {
            arr2[i] = scanner.nextInt();
        }
        // ... (rest of the code) ...
    }
}
```

Output

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Part 2:

Progress: 100%

Details:

Direct link to the file in the homework related branch from Github (should end in `.java`)

URL #1

<https://github.com/AlCSS442/asa9-IT114-1003M2-Homework/M2/Problem3.java>



URL

<https://github.com/AlCSS442/asa>

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Part 3:

Progress: 100%

Details:

Briefly explain `how` the code solves the challenges (note: this isn't the same as `what` the code does)

Your Response:

I solved the first challenge by iterating through each array and converting all values to their positive version. Since the arrays contain multiple data types, I checked each element's type and applied the appropriate conversion using casting. For string representations of numbers, I parsed them into numeric types, made them positive, and then converted them back to their original form. The main thing was making sure that every value becomes positive, BUT keeps the array's original data type.

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Section #4: (2 pts.) Problem 4 - Strings

Progress: 100%

≡ Task #1 (2 pts.) - Edit the `transformText` method to solve the challenges

Progress: 100%

Details:

- Only make edits where noted via provided comments
- Challenge 1: Remove non-alphanumeric characters except spaces
- Challenge 2: Convert text to Title Case
- Challenge 3: Trim leading/trailing spaces and remove duplicate spaces
- Result 1-3: Assign final phrase to `placeholderForModifiedPhrase`
- Step 1: sketch out plan using comments (include ucid and date)
- Step 2: Add/commit your outline of comments (required for full credit)
- Step 3: Add code to solve the problem (add/commit as needed)

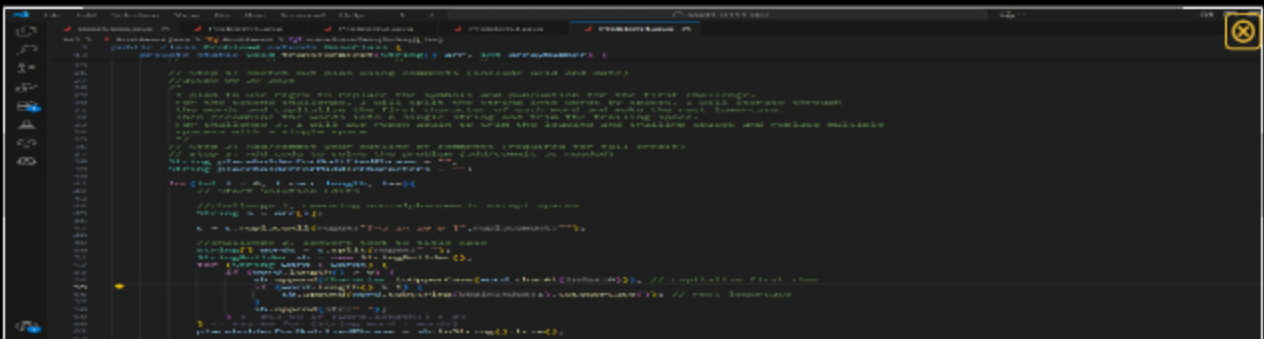
Part 1:

Progress: 100%

Details:

Two screenshots are expected

1. Snippet of relevant code showing solution (with ucid/date comment)
2. Full output of executing the program



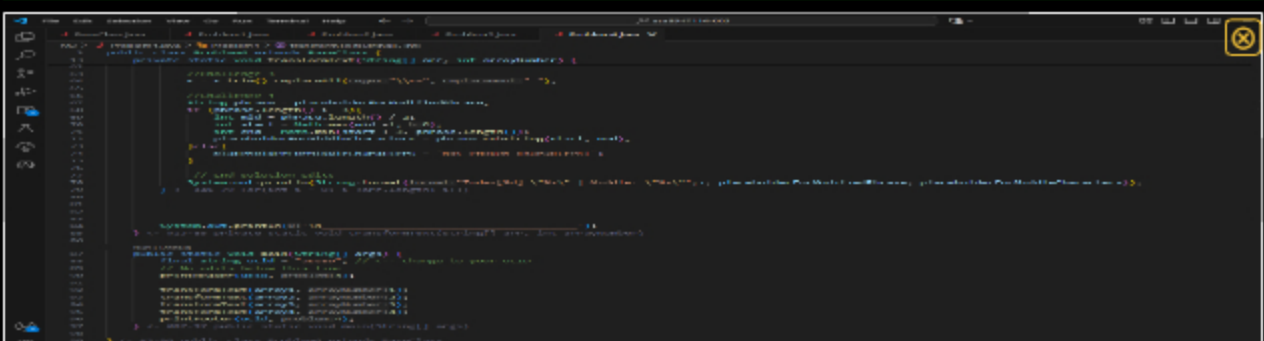
```
// CHALLENGE 1: Remove non-alphanumeric characters except spaces
// CHALLENGE 2: Convert text to Title Case
// CHALLENGE 3: Trim leading/trailing spaces and remove duplicate spaces

// Step 1: sketch out plan using comments (include ucid and date)
// Step 2: Add/commit your outline of comments (required for full credit)
// Step 3: Add code to solve the problem (add/commit as needed)

// CHALLENGE 1: Remove non-alphanumeric characters except spaces
// CHALLENGE 2: Convert text to Title Case
// CHALLENGE 3: Trim leading/trailing spaces and remove duplicate spaces

// Step 1: sketch out plan using comments (include ucid and date)
// Step 2: Add/commit your outline of comments (required for full credit)
// Step 3: Add code to solve the problem (add/commit as needed)
```

Part 1 of code



```
// CHALLENGE 1: Remove non-alphanumeric characters except spaces
// CHALLENGE 2: Convert text to Title Case
// CHALLENGE 3: Trim leading/trailing spaces and remove duplicate spaces

// Step 1: sketch out plan using comments (include ucid and date)
// Step 2: Add/commit your outline of comments (required for full credit)
// Step 3: Add code to solve the problem (add/commit as needed)
```



```

import java.util.*;

public class Solution {
    public String transformText(String input) {
        // Challenge 1: Remove non-alphanumeric characters except spaces
        String cleaned = input.replaceAll("[^a-zA-Z0-9 ]", "");

        // Challenge 2: Convert each word to title case
        String[] words = cleaned.split(" ");
        for (int i = 0; i < words.length; i++) {
            words[i] = words[i].toLowerCase().replaceFirst(
                "[a-z]", words[i].toUpperCase().charAt(0) + words[i].toLowerCase().substring(1)
            );
        }

        // Challenge 3: Trim leading and trailing spaces and replace multiple spaces with a single space
        String result = String.join(" ", words);
        result = result.trim();

        return result;
    }
}

```

Execution



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Part 2:

Progress: 100%

Details:

Direct link to the file in the homework related branch from Github (should end in `.java`)

URL #1

<https://github.com/AICSS442/asa89-IT114-1003M2-Homework/M2/Problem4.java>


URL

<https://github.com/AICSS442/asa>


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Part 3:

Progress: 100%

Details:

Briefly explain `how` the code solves the challenges (note: this isn't the same as `what` the code does)

Your Response:

I solved the first challenge by removing all non-alphanumeric characters except spaces using regex. For the second challenge, I converted each word to title case by capitalizing the first letter and making the rest lowercase. I also trimmed leading and trailing spaces and replaced multiple spaces with a single space to clean up the text. Essentially, it involved some loops and regex.



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Task #2 (+ 1.11 pts.) - Edit the `transformText` method to solve the extra credit challenge (challenge 4)

Progress: 100%

Details:

- Only make edits where noted via provided comments
- Challenge 4: Extract middle 3 characters (beginning starts at middle of phrase)
- Assign result to 'placeholderForMiddleCharacters'
- If not enough characters assign "Not enough characters"
- Step 1: sketch out plan using comments (include uuid and date)
- Step 2: Add/commit your outline of comments (required for full credit)
- Step 3: Add code to solve the problem (add/commit as needed)

Part 1:

Progress: 100%

Details:

Two screenshots are expected

1. Snippet of relevant code showing solution (with uuid/date comment)
2. Full output of executing the program

Challenge 4

Execution

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Part 2:

Progress: 100%

Details:

Briefly explain how the code solves the extra credit challenge (note: this isn't the same as

briefly explain how the code solves the extra credit challenge (note: this isn't the same as what the code does)

Your Response:

This was located in the loop. I first found the middle of the string. From there, it selected the 3 chracters around it. If the phrase was too short, it let gave a message to let the user know.

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Section #5: (2 pts.) Misc

Progress: 100%

Task #1 (0.67 pts.) - Github Details

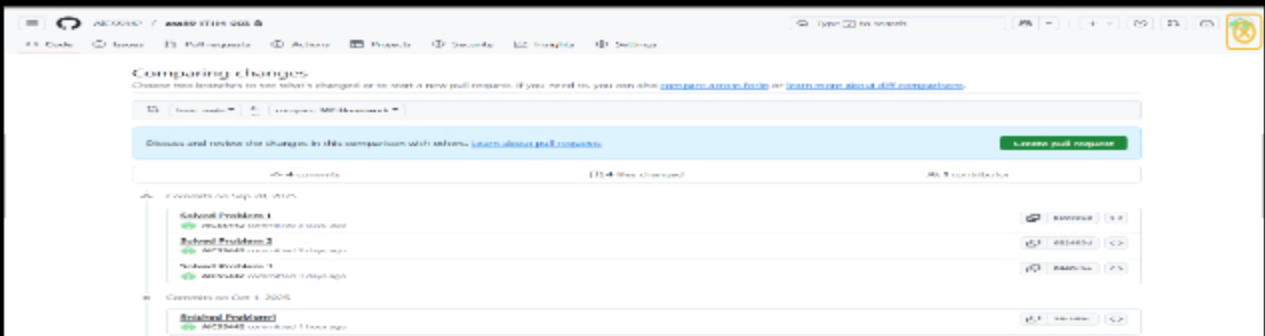
Progress: 100%

Part 1:

Progress: 100%

Details:

From the Commits tab of the Pull Request screenshot the commit history Following minimum should be present



Commit history

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Part 2:

Progress: 100%

Details:

Include the link to the Pull Request (should end in `/pull/#`)

URL #1

[https://github.com/AlCSS442/asa89-](https://github.com/AlCSS442/asa89-IT114003/)

[IT114003/](https://github.com/AlCSS442/asa89-IT114003/)



URL

[https://github.com/AlCSS442/asa89-](https://github.com/AlCSS442/asa89-IT114003/)

Task #2 (0.67 pts.) - WakaTime - Activity

Progress: 100%

Details:

- Visit the WakaTime.com Dashboard
- Click **Projects** and find your repository
- Capture the overall time at the top that includes the repository name
- Capture the individual time at the bottom that includes the file time
- Note: The duration isn't relevant for the grade and the visual graphs aren't necessary

Projects • asa89-IT114-003

5 hrs 41 mins over the Last 7 Days in asa89-IT114-003 under all branches. 📌

waketime

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Task #3 (0.67 pts.) - Reflection

Progress: 100%

Task #1 (0.33 pts.) - What did you learn?

Progress: 100%

Details:

Briefly answer the question (at least a few decent sentences)

Your Response:

I learned about some concepts, specifically ArrayList vs arrays, instanceof checks, and casting. Arrays are fixed in size, but ArrayLists are dynamic and only work with objects. instanceof lets us find out the type of an object at runtime. Casting is important because it explicitly converts one data type to another so the program knows how to properly use that data.

⇒ Task #2 (0.33 pts.) - What was the easiest part of the assignment?

Progress: 100%

Details:

Briefly answer the question (at least a few decent sentences)

Your Response:

I'd say the first question of the homework was the simplest to figure out. It was a good introduction to the assignment, as the questions increased in difficulty.

⇒ Task #3 (0.33 pts.) - What was the hardest part of the assignment?

Progress: 100%

Details:

Briefly answer the question (at least a few decent sentences)

Your Response:

The hardest part I would say would probably be Problem 4. It really required me to think outside of the box, especially as it related to the Challenge question. I also needed a bit of guidance for Challenge 2 of Question 4.